

CAPSTONE · INTERNSHIP REPORT

JanusScale

Event-Driven Kubernetes Autoscaling with Multivariate Anomaly
Detection

Statistical process-control framework for telemetry-driven scaling decisions

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Single-metric scaling can miss correlated failure signatures.



“Can runtime telemetry be transformed into a statistically grounded, workload-aware scaling decision?”



Objectives & Contributions

01

INGEST

Collect and process
Kubernetes runtime
telemetry.

02

DETECT

6D representation +
MEWMA / Hotelling T^2 .

03

ATTRIBUTE

Connect anomalous
dimensions to workload
dependencies.

04

DECIDE

Select replica target under
runtime constraints and
safety policies.

05

ENFORCE

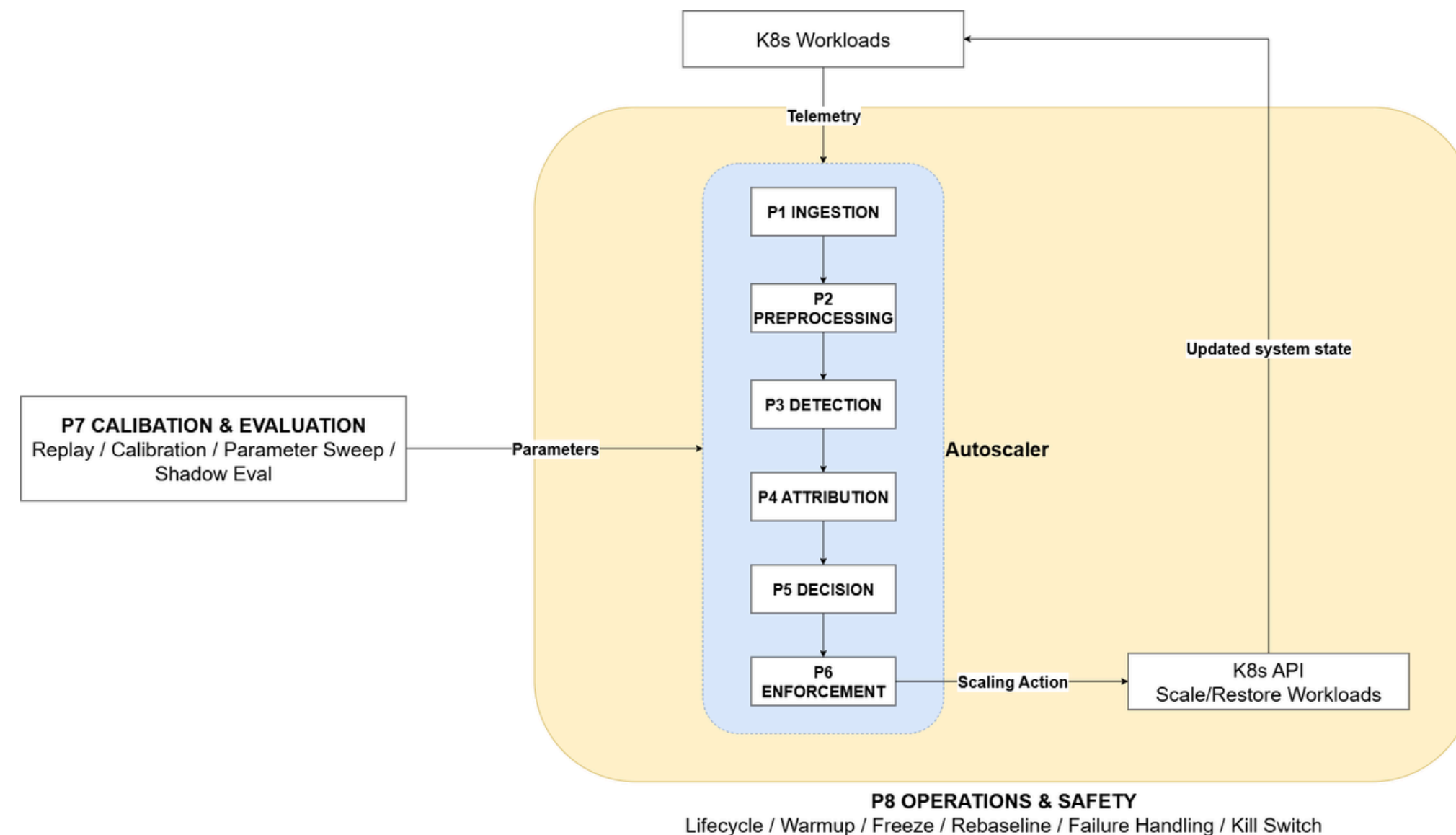
Apply scaling through a
dedicated Kubernetes
actuator.

CORE CONTRIBUTION

An end-to-end event-driven pipeline connecting telemetry → analysis → attribution → decision → enforcement

PART 2 — HOW

JanusScale Architecture



P1 Ingestion

P2 Preprocessing

P3 Detection

P4 Attribution

P5 Decision

P6 Enforcement

P7 — Calibration & Evaluation
supports offline / replay evaluation.

Figure 4.1 — JanusScale Overall System Architecture

P1–P6 form the online control loop. P7 supports evaluation; P8 provides operational and safety controls.

PART 2 — HOW

From Raw Telemetry to a Six-Dimensional Observation

cAdvisor

- CPU
- RAM

Hubble

- Request rate
- Error rate
- Request-duration signals

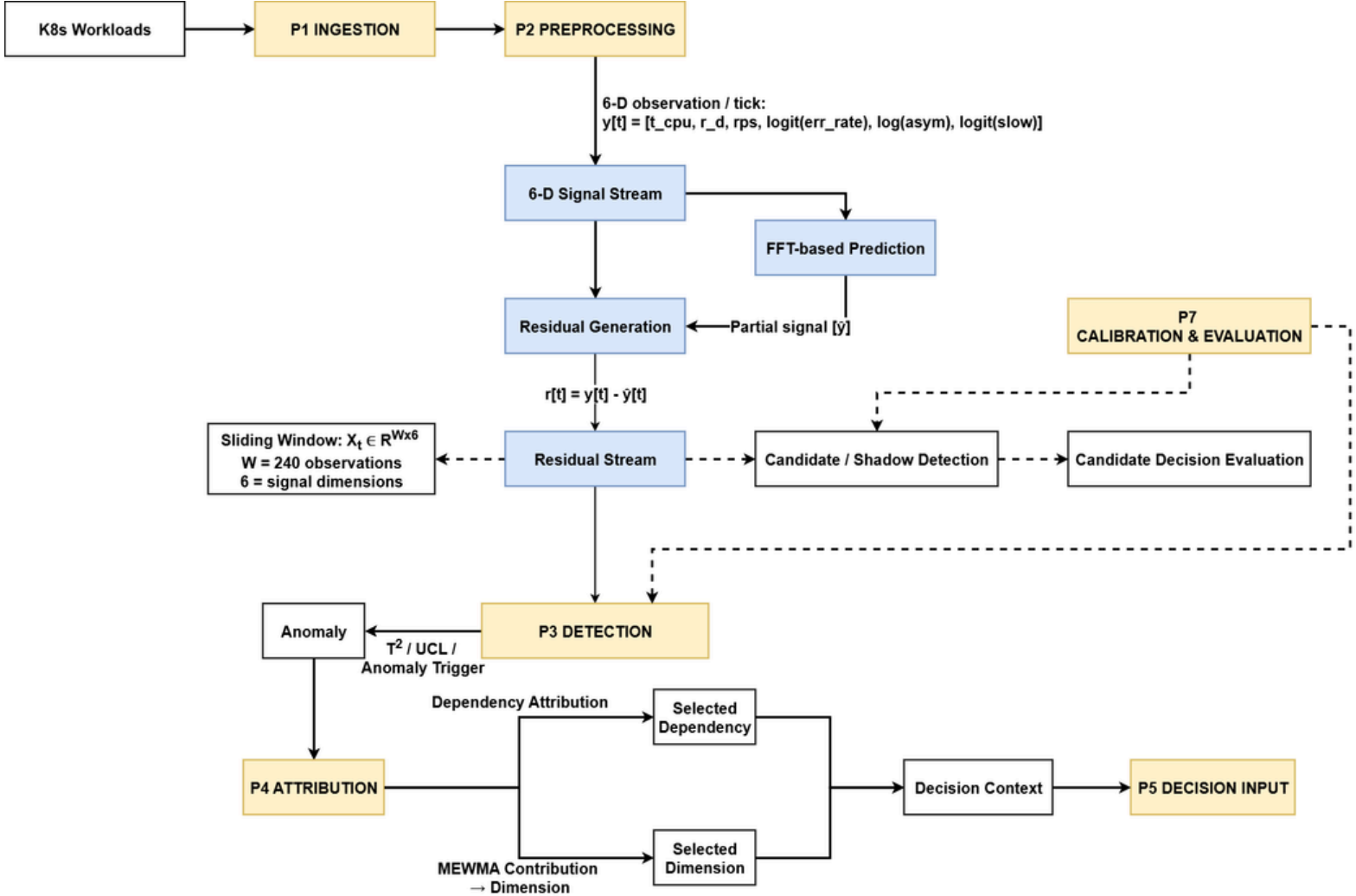
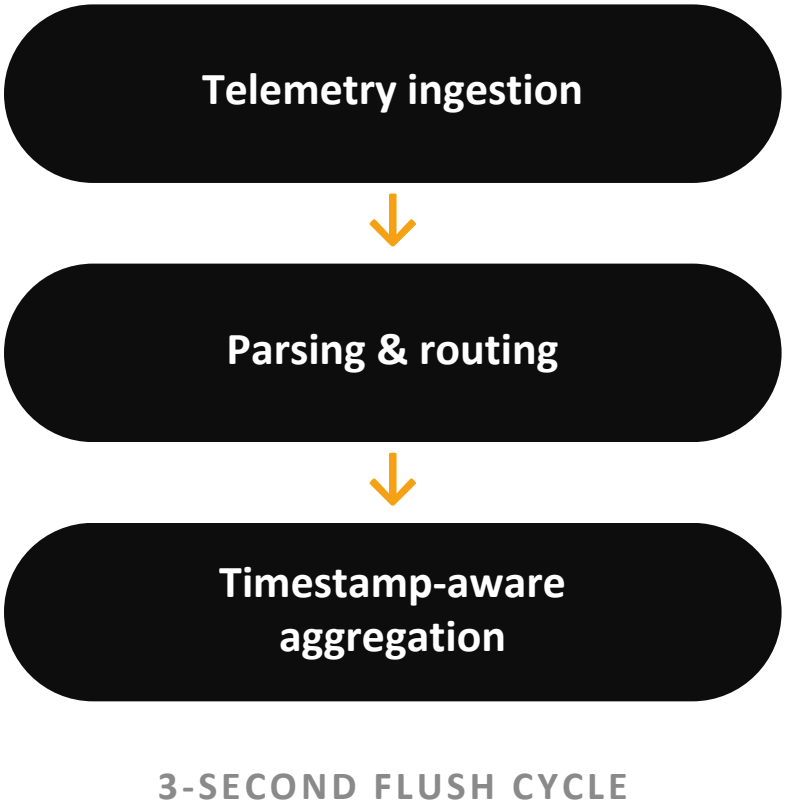


Figure 4.2 — Telemetry-to-Decision Data Flow

6D OBSERVATION: CPU · RAM · RPS · ERROR · ASYM · SLOW

ASYM is a derived / aggregated signal, not a single raw metric.

Multivariate Detection & Technical Views

01 MEWMA / Hotelling T^2

PRIMARY ANOMALY DETECTOR

$$\begin{aligned} z_t &= \lambda x_t + (1 - \lambda) z_{t-1} \\ T^2_t &= z_t^T \Sigma z^{-1} z_t \\ T^2_t > UCL &\Rightarrow \text{anomaly} \end{aligned}$$

This is the primary anomaly detector.

02 RFFT / FFT

TEMPORAL EXPLANATION

$$\begin{aligned} X[k] &= \sum x[n] \cdot e^{(-j2\pi kn/N)} \\ A[k] &= |X[k]| \end{aligned}$$

Applied to CPU · RPS · ASYM · SLOW. RAM · ERROR remain direct signals.

03 GARCH / DCC

VOLATILITY & DEPENDENCY CONTEXT

$$\begin{aligned} \sigma^2_t &= \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1} \\ Q_t &= (1-a-b) \bar{Q} + a \varepsilon_{t-1} \varepsilon^T_{t-1} + b Q_{t-1} \end{aligned}$$

Time-varying variance + dependency context.

MEWMA / T^2 = detection RFFT = temporal explanation GARCH / DCC = supporting context

From Anomaly to a Workload-Specific Action

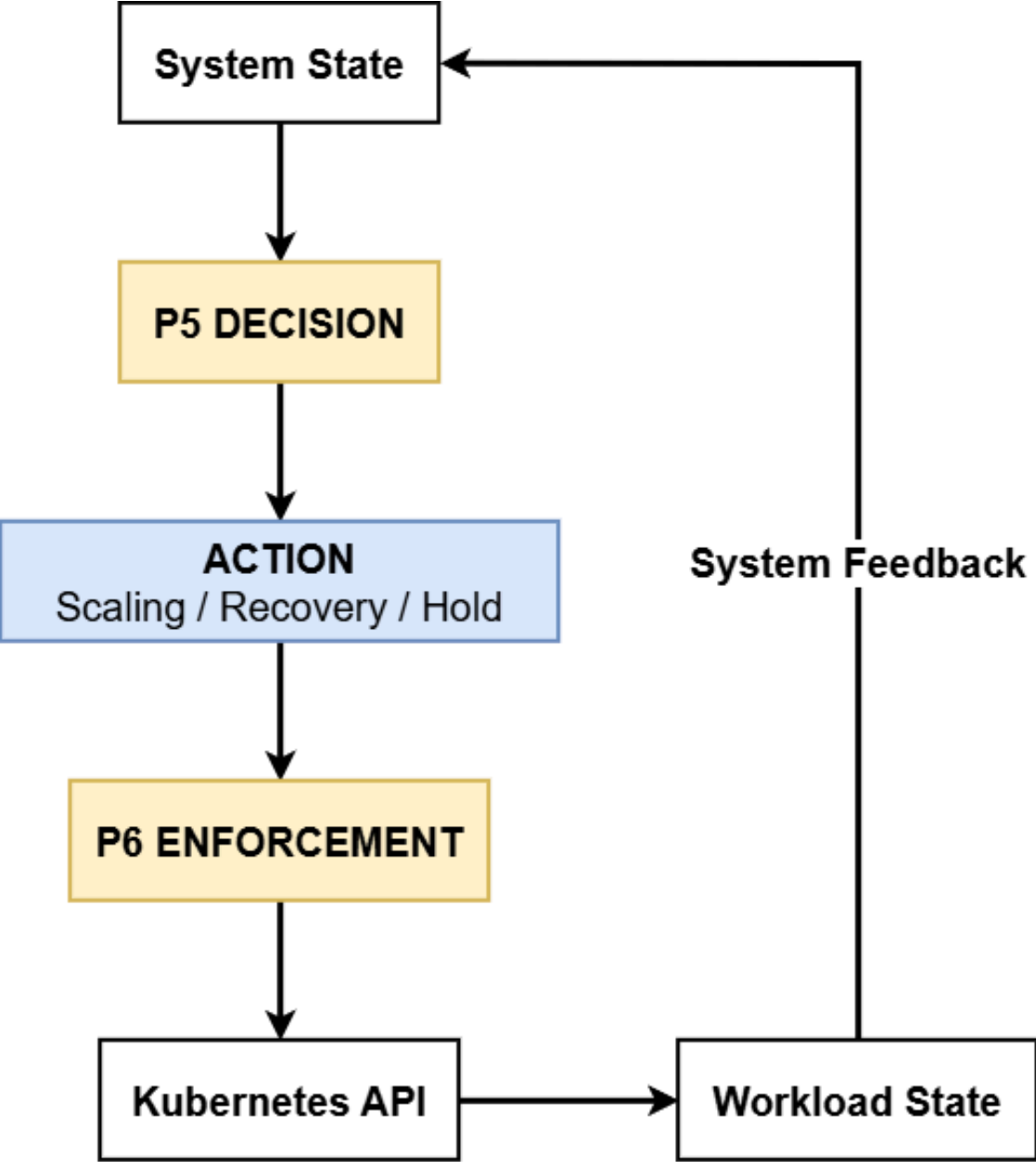
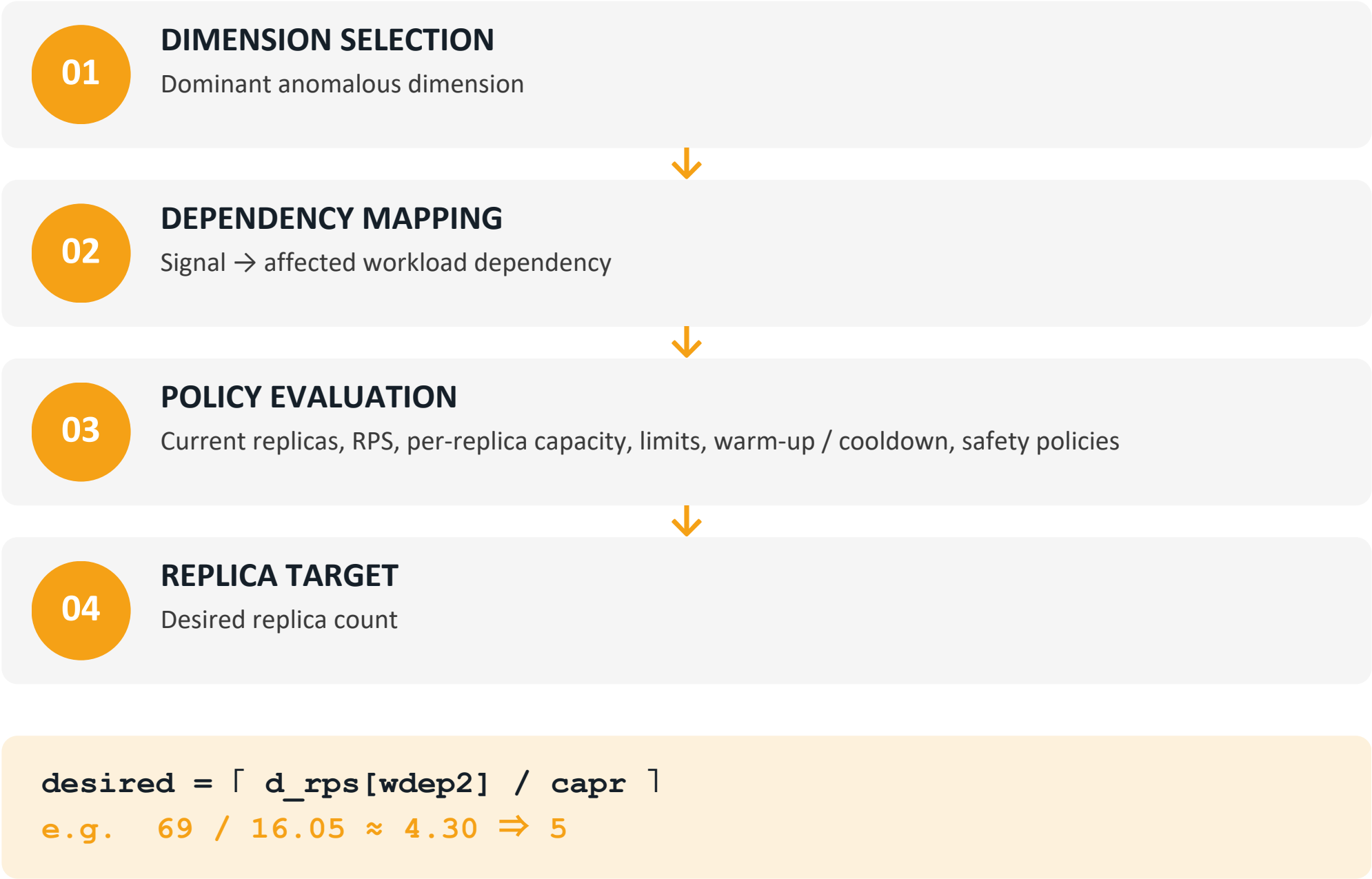
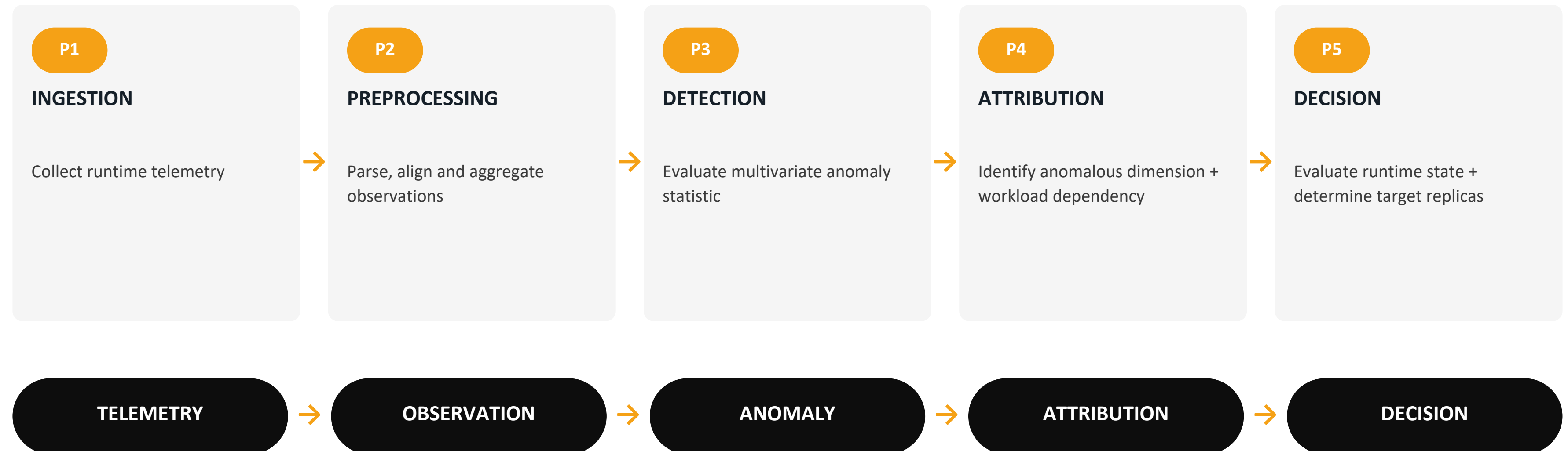


Figure 4.3 — Decision & Recovery Flow

Attribution identifies what changed — decision determines how much capacity is required.

From Telemetry to Decision



Runtime data flow — architecture detail lives on the System Architecture slide.

Decision & Enforcement

P5 · DECISION

- Current replicas
- RPS
- Per-replica capacity
- Resource limits
- Warm-up
- Cooldown
- Freeze / safety constraints



Desired replica count

P6 · ENFORCEMENT

Authenticated request



Target workload whitelist



Per-deployment locking



Kubernetes scaling



Post-action verification



JSONL audit event

Analytical decision and direct Kubernetes modification are separated by an explicit control boundary.

Calibration & Evaluation

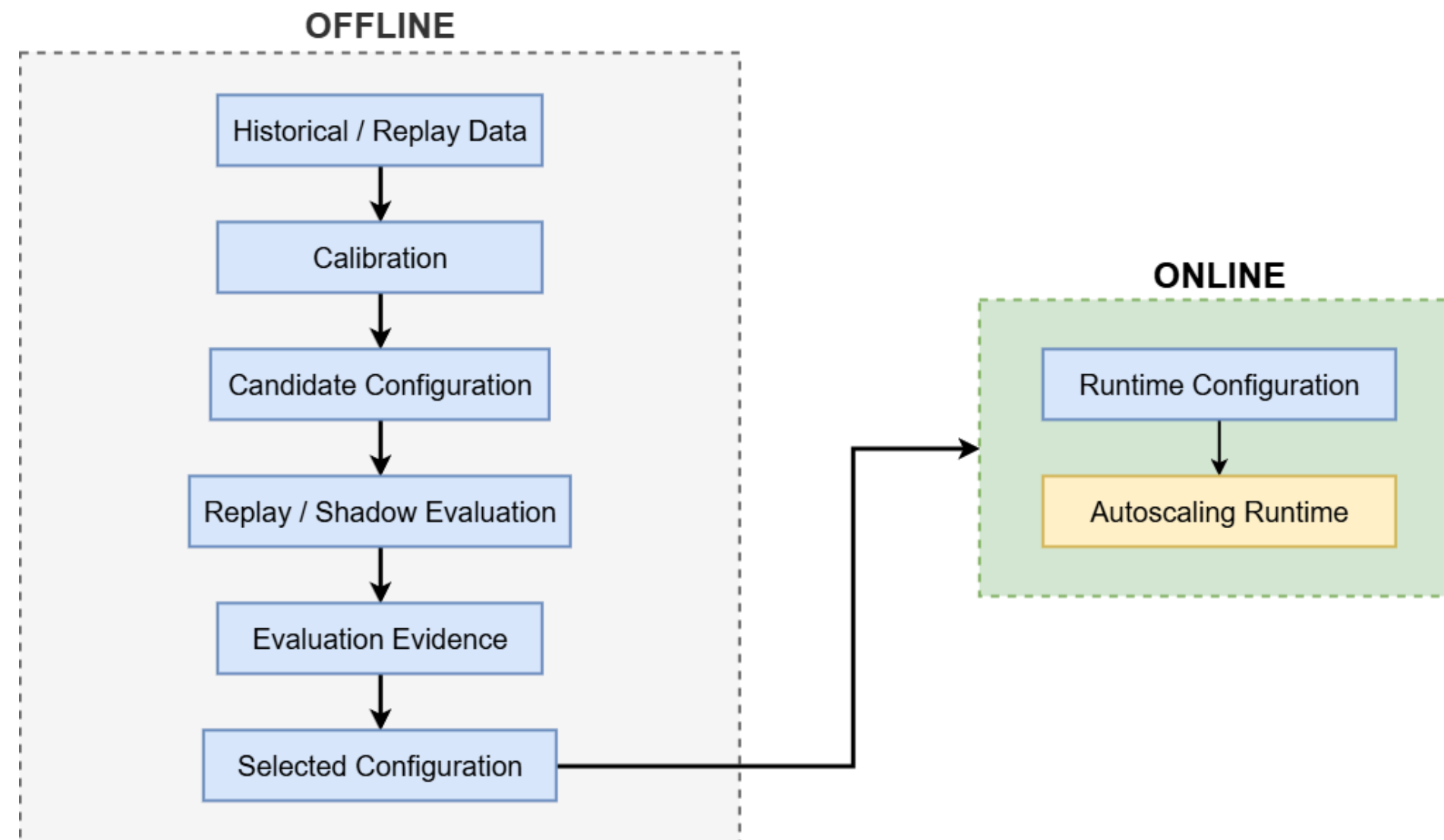
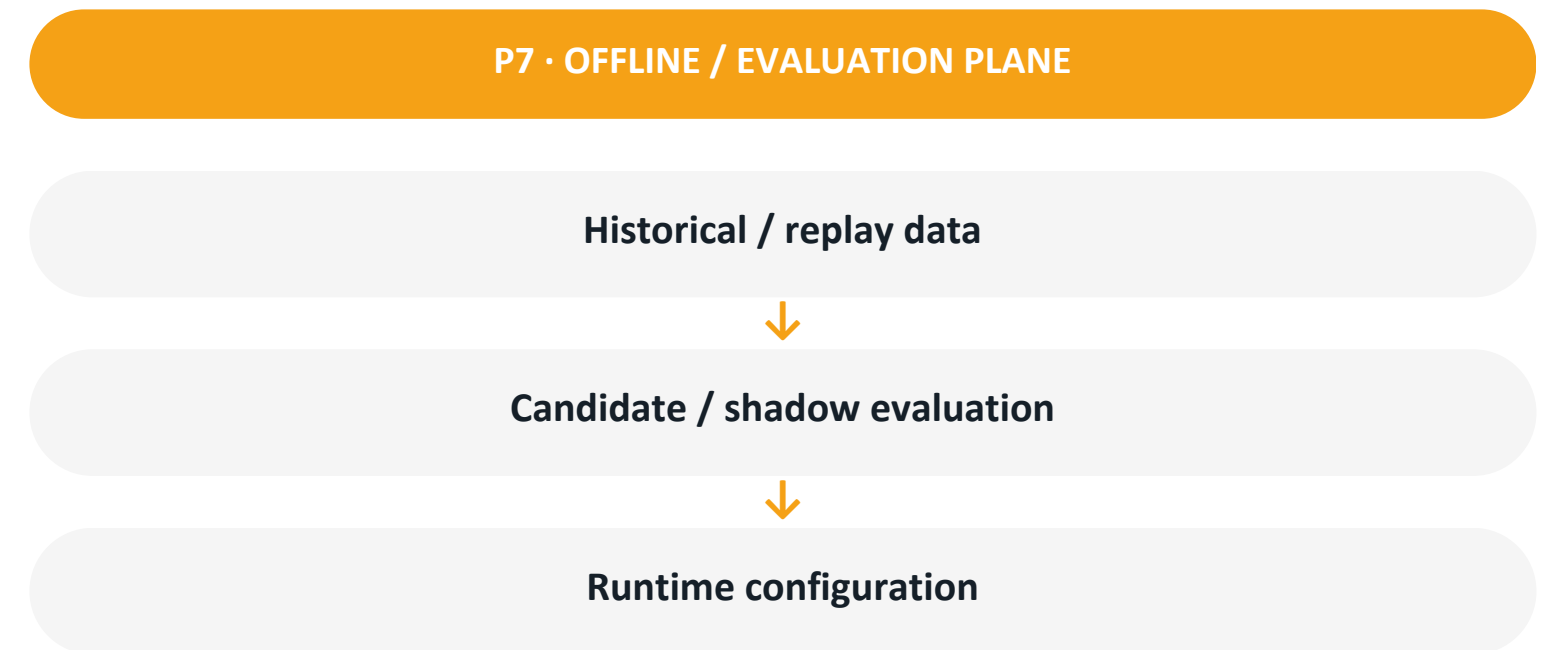


Figure 4.4 — Calibration & Evaluation Flow

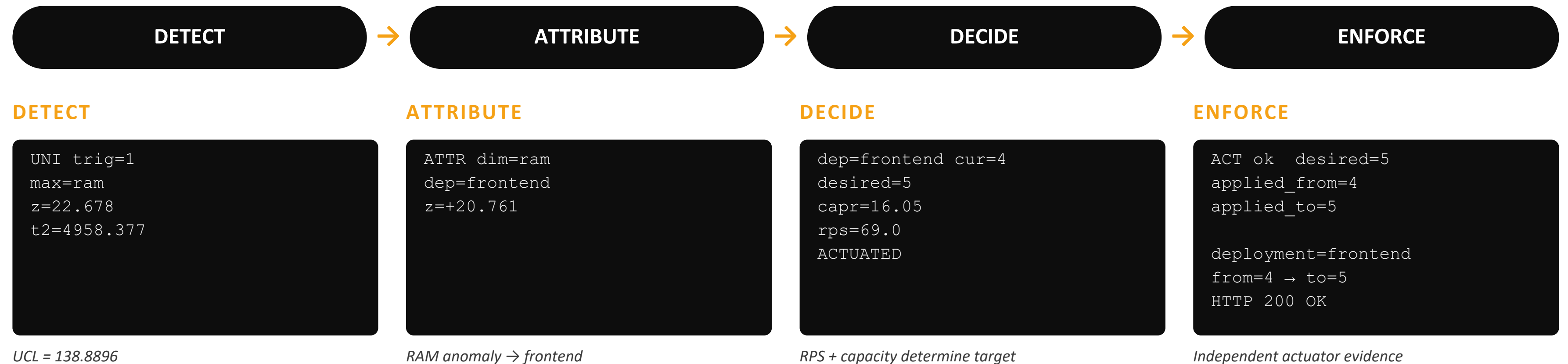


PURPOSE

- Evaluate detector behavior
- Tune / calibrate configuration
- Support replay and shadow analysis
- Provide evidence without becoming another online control stage

P7 supports the online control loop; it does not replace the runtime decision path.

Verified Closed-Loop Actuation



The anomaly was RAM; replica sizing used workload RPS and per-replica capacity.

Implementation Evidence & Current Limits

VERIFIED

Multivariate detection

T² anomaly event directly recorded.

Attribution

RAM anomaly mapped to frontend.

Decision

Current replicas, RPS, capacity and desired replicas recorded.

Actuation

Independent actuator evidence confirms frontend: 4 → 5.

Calibration

Calibration/shadow evaluation separated from production runtime configuration.

CURRENT LIMITS

1

Only one complete attribution event is directly documented in supplied runtime evidence.

2

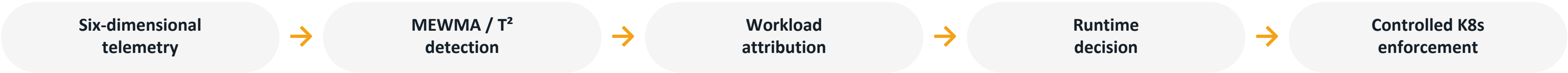
The report does not establish globally optimal autoscaling or long-term production performance.

3

Independent Kubernetes audit/history evidence for the preceding 5 → 4 transition is unavailable.

Actuator logs prove applied scaling actions, but do not by themselves establish detector → attribution → decision causality for every action.

Conclusion & Next Steps



KEY OUTCOMES

- ✓ Multivariate telemetry rather than single-metric observation
- ✓ MEWMA / Hotelling T² as the primary anomaly detector
- ✓ Explicit separation of detection, attribution, decision and enforcement
- ✓ Controlled actuation with authentication and workload restrictions
- ✓ Runtime safeguards around scaling actions
- ✓ Directly evidenced closed-loop scale-out

NEXT STEPS

01

Evidence alignment
Align detector, decision, Kubernetes-state and audit timestamps.

02

Broader runtime evidence
Collect more verified attribution events across dimensions and workloads.

03

Longer-run evaluation
Expand evaluation toward workload behavior, scaling response and recovery evidence.